

SMART CAMPUS EVENT MANAGEMENT SYSTEM

School of Computing

**Bachelor of Technology
in
Computer Science & Engineering**

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May, 2026

BONAFIDE CERTIFICATE

It is certified that the work contained in the Full Stack Application Development report titled "SMART CAMPUS EVENT MANAGEMENT SYSTEM" by B.GOWTHAM KUMAR(VTU26616) has been carried out in Winter semester of Academic year 2025-2026(WS2526).

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DECLARATION

I declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. I also declare that i have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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ABSTRACT

The Smart Campus Event Management System is designed to simplify and improve how college events are organized and managed. It provides a user-friendly platform where students can easily explore events, register online, and keep track of their participation. At the same time, administrators can create events, manage registrations, and monitor overall activity through an interactive dashboard. The system integrates modern features such as secure login, OTP-based verification, email notifications, and real-time updates, making the process more efficient and reliable. By reducing manual work and paperwork, it ensures better coordination and communication between students and organizers. Overall, this system creates a more connected campus environment by making event participation seamless, organized, and accessible for everyone.

Keywords: Smart Campus, Event Management System, Student Portal, Admin Dashboard, OTP Verification, Email Notification, Spring Boot, Web Application.

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LIST OF ACRONYMS AND ABBREVIATIONS

UI	User Interface
UX	User Experience
DB	Database
RDBMS	Relational Database Management System
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
JPA	Java Persistence API
API	Application Programming Interface
MVC	Model View Controller
SMTP	Simple Mail Transfer Protocol
CRUD	Create, Read, Update, Delete
DTO	Data Transfer Object

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Chapter 1

INTRODUCTION

1.1 Introduction

In today's digital era, managing college events manually can be time-consuming, unorganized, and inefficient. Students often face difficulties in staying updated about upcoming events, while administrators struggle with handling registrations, announcements, and coordination. To address these challenges, the Smart Campus Event Management System is developed as a centralized platform that simplifies the entire process of event management within a campus

This system provides an easy-to-use interface where students can browse events, register online, and track their participation, while administrators can efficiently create events, manage registrations, and monitor activities through a dashboard. By integrating features such as secure authentication, OTP verification, and email notifications, the system enhances communication and reduces manual effort, making event management more streamlined, reliable, and accessible.

1.2 Aim of the Project

The aim of this project is to develop a Smart Campus Event Management System that makes organizing and participating in college events simple and efficient. It focuses on providing a centralized platform where students can easily explore and register for events, while administrators can manage event details, registrations, and announcements without manual effort. By incorporating features like secure login, OTP verification, and real-time updates, the system aims to improve communication, reduce paperwork, and create a smooth, organized experience for both students and event organizers.

1.3 Scope of the Project

The scope of this project covers the design and development of a Smart Campus Event Management System that streamlines event-related activities within a college environment. It includes features such as user registration and authentication, event creation and management, online event registration, and tracking of participation. The system supports both student and admin roles, allowing students to browse and register for events, while administrators can manage events, announcements, and monitor registrations through a centralized dashboard.

Additionally, the project incorporates functionalities like OTP-based verification, email notifications, and real-time updates to enhance user experience and communication. The system is scalable and can be further extended with advanced features such as payment integration for paid events, analytics dashboards, and integration with other campus systems, making it adaptable for future enhancements.

1.4 Framework

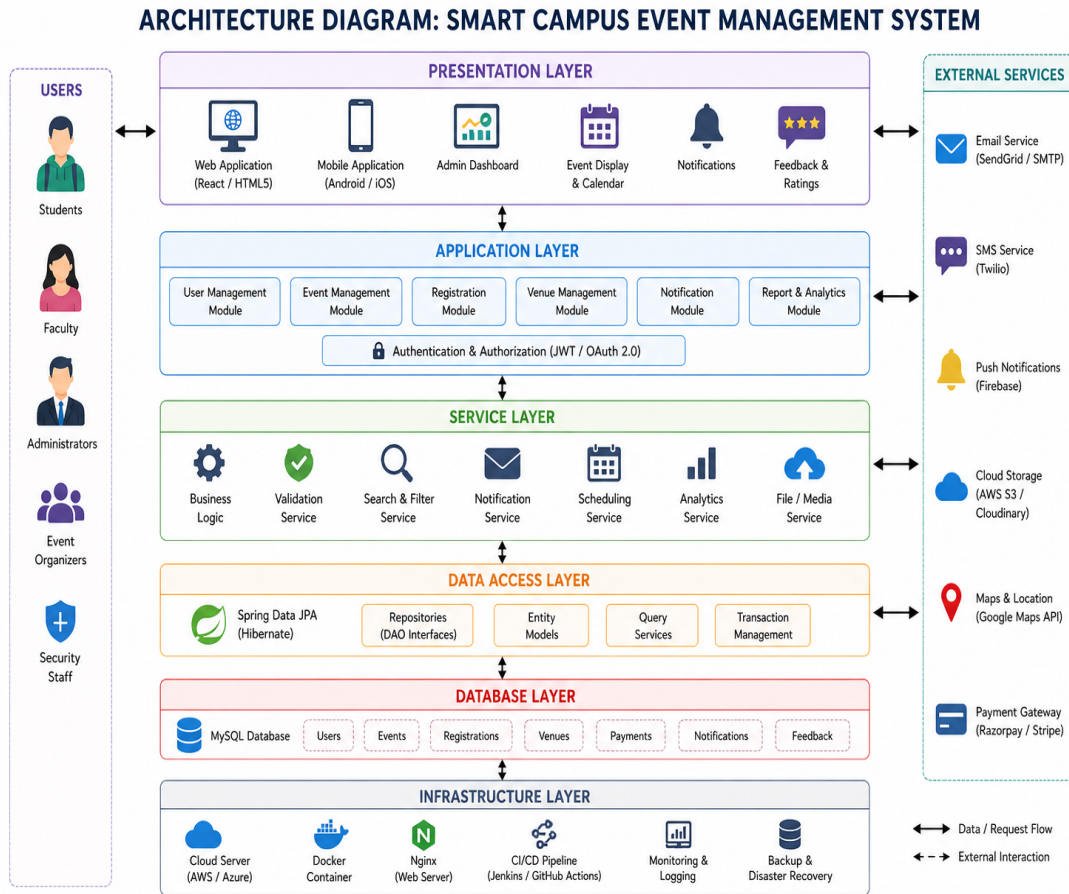


Figure 1.1: Framework

Figure 1.1 shows the layered framework of a Smart Campus Event Management System, including users, presentation layer, application modules, service layer, data layer, and infrastructure. It illustrates how students, faculty, and administrators interact with the system to manage events, registrations, and notifications. It also highlights the integration of external services and the interactions between different functional components within the system.

Chapter 2

SURVEY OF SIMILAR APPLICATION IN MARKET

In today's digital world, several event management applications are already available in the market that help organizations plan and manage events efficiently. Platforms like Cvent, Whova, and Eventbrite are widely used for organizing conferences, seminars, and large-scale events. These systems provide features such as online registration, ticketing, event promotion, attendee engagement, and analytics. Most modern platforms also support virtual and hybrid events, making them flexible and scalable for different types of users.

For educational institutions, specialized tools like Eventleaf and Zoho Backstage are used to manage campus events, workshops, and student activities. These systems focus on simplifying scheduling, improving communication, and centralizing event data across departments. However, many existing solutions are either costly, complex, or not fully customized for college environments. This creates a need for a simplified and cost-effective Smart Campus Event Management System that specifically addresses student and administrator requirements within an educational institution.

Chapter 3

FRAMEWORK DESCRIPTION

3.1 Frontend Implementation

3.1.1 Environment Setup

The frontend environment for the Smart Campus Event Management System is set up using basic web technologies such as HTML, CSS, and JavaScript, along with the Thymeleaf template engine integrated into Spring Boot. Development is carried out using Visual Studio Code, with necessary extensions for HTML, CSS, Java, and Spring Boot support. A web browser such as Google Chrome is used for testing and debugging the user interface. The project follows a structured folder organization with separate directories for templates and static resources like CSS and JavaScript. Version control can be managed using Git, ensuring smooth development, collaboration, and tracking of changes throughout the project lifecycle.

3.1.2 Structure of the Project

The Smart Campus Event Management System is organized using a layered architecture to ensure clarity, scalability, and ease of maintenance. The project is divided into multiple packages based on functionality.

The controller layer handles user requests and navigation between pages, while the service layer contains the core business logic of the application. The repository layer manages database interactions using Spring Data JPA, and the model layer defines the entity classes that represent database tables such as users, events, and registrations.

Additionally, the project includes DTO (Data Transfer Object) classes for structured data handling and an exception package for managing errors effectively. The config package is used for security and application configuration. The frontend is placed in the resources folder, which contains templates (HTML pages using Thymeleaf) and static files (CSS and JavaScript). This well-defined structure ensures smooth communication between components and makes the system easy to develop, test, and extend.

3.1.3 Execution Procedure

The Smart Campus Event Management System is executed by first setting up the development environment and configuring the application properties, including database, email, and OTP services. The project is then built and run using Spring Boot through an IDE like Visual Studio Code or IntelliJ. Once the server starts, the application can be accessed through a web browser using the default localhost port. Users can register, log in, and interact with the system based on their roles, while administrators manage events and monitor activities. The system processes requests through controllers, executes business logic in services, and stores or retrieves data from the database seamlessly.

3.2 Backend Implementation

3.2.1 Environment Setup

The backend environment for the Smart Campus Event Management System is set up using Java and Spring Boot. Development is carried out in an IDE such as IntelliJ IDEA or Visual Studio Code with necessary Java and Spring extensions. Maven is used for dependency management and project build. A relational database like MySQL is configured for data storage. Application properties are set for database connection, email, and OTP services, ensuring smooth backend processing and integration.

3.2.2 Structure of the Project

The backend of the Smart Campus Event Management System follows a well-organized layered structure to ensure efficient development and maintainability. It is divided into packages such as controller, service, repository, and model. The controller layer manages incoming requests and responses, while the service layer handles the core business logic. The repository layer uses Spring Data JPA to interact with the database, and the model layer defines the entity classes representing tables like users, events, and registrations.

3.2.3 Execution Procedure

The backend of the Smart Campus Event Management System is executed by configuring the database, application properties, and required dependencies. The project is built and run using Spring Boot through an IDE like IntelliJ or Visual Studio Code. Once the server starts, APIs and controllers handle incoming requests from the frontend. The service layer processes the logic, and the repository layer interacts with the database to store and retrieve data efficiently.

3.3 Database Implementation

3.3.1 Database Configuration

The database is configured using MySQL to store and manage application data efficiently. Tables such as users, events, registrations, and announcements are created with appropriate fields and constraints. The database connection is established through Spring Boot configuration using application properties, ensuring secure access, smooth data retrieval, and reliable storage for system operations.

3.1.2 Schema Structure

The database schema is designed to organize data into structured tables such as users, events, registrations, feedback, and announcements. Each table contains relevant attributes with primary and foreign keys to maintain relationships. This relational structure ensures data integrity, avoids redundancy, and enables efficient querying, updating, and management of event-related information within the system.

3.1.3 Tables created

- Users
- Events
- Registrations
- Feedbacks
- Announcements
- Departments
- Roles

Chapter 4

METHODOLOGIES

4.1 Project Architecture

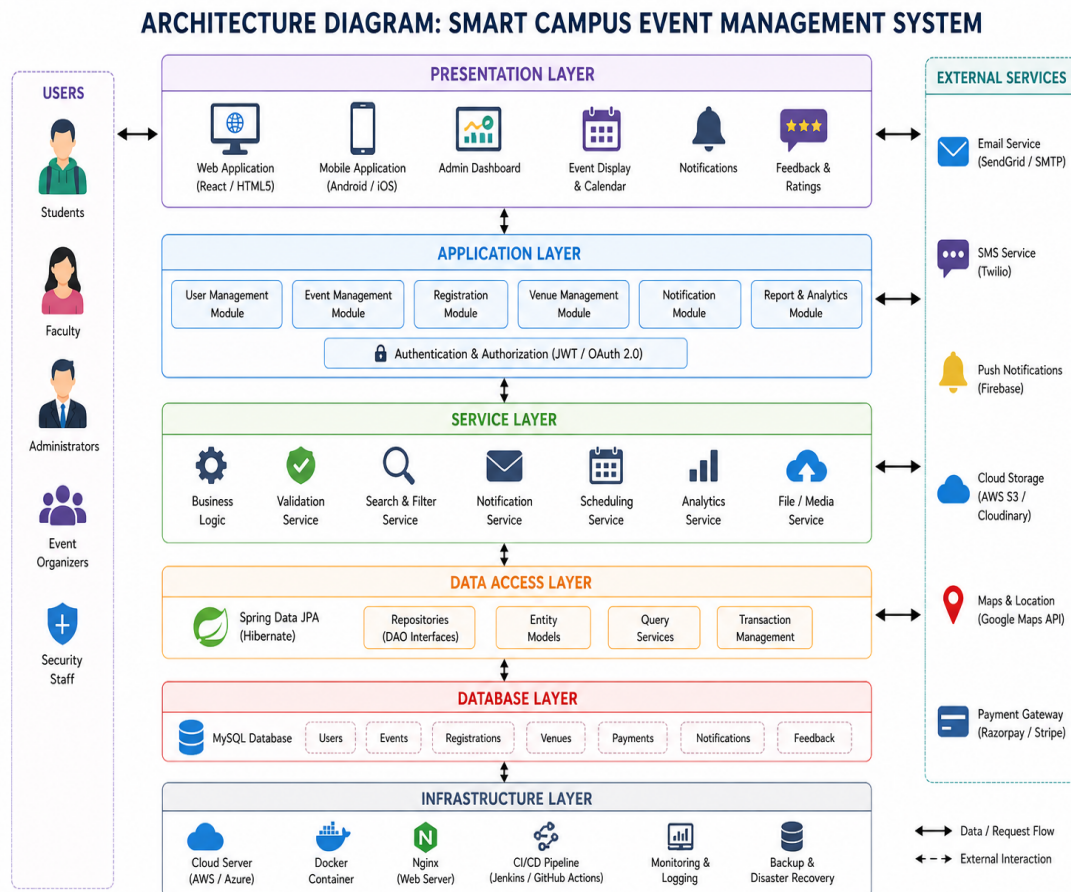


Figure 4.1: Architecture diagram

Figure 4.1 shows the architecture of the Smart Campus Event Management System with presentation, application, service, data access, database, and infrastructure

layers. It explains how users interact with the system through web and mobile interfaces to manage events, registrations, and notifications. The diagram also highlights integration with external services and the flow of data between different components of the system.

4.2 DataFlow Diagram

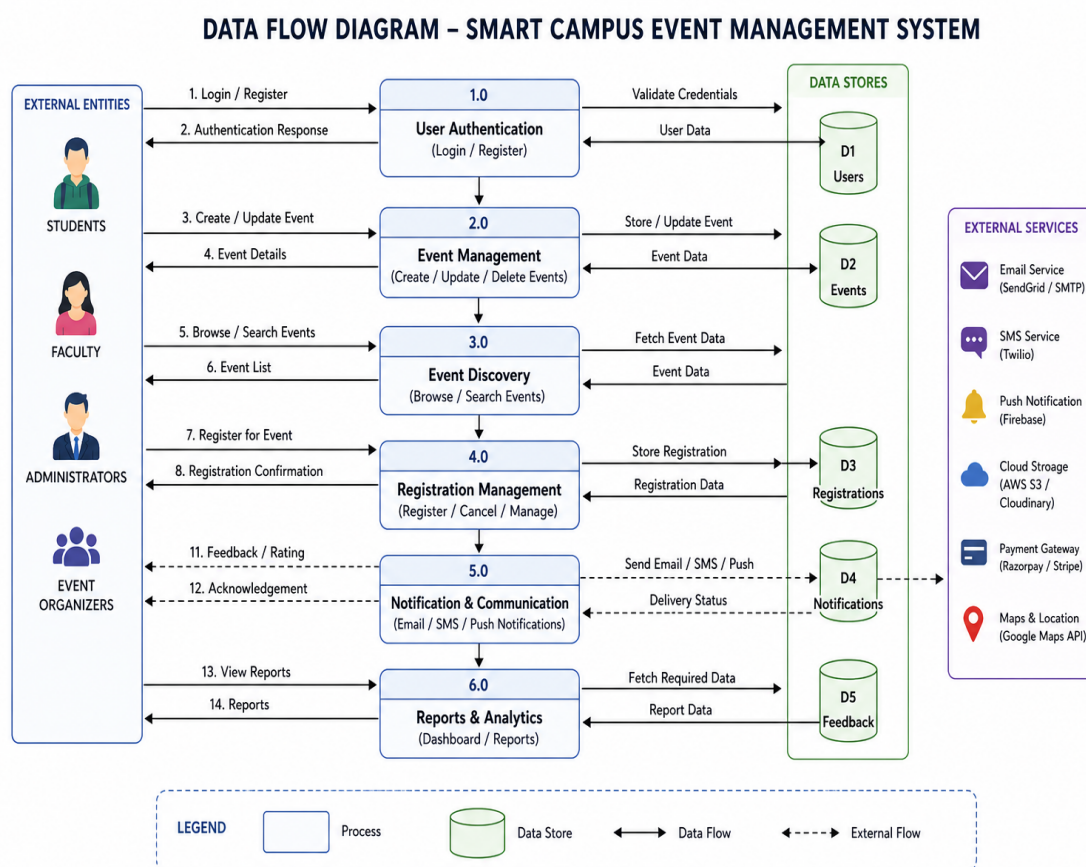


Figure 4.2: DataFlow Diagram

Figure 4.2 illustrates the data flow of the Smart Campus Event Management System, showing how data moves between users, processes, and data stores. It explains key operations such as user authentication, event management, event discovery, registration, notifications, and report generation. The diagram highlights how informa-

tion is stored and retrieved from databases and how external services are used for communication and other functionalities.

4.3 User Authentication and Registration:

This module handles user signup, login, and role-based access for students and administrators. It allows new users to register by providing basic details, verifies their identity using OTP, and securely stores credentials in the database. The module also manages login authentication, session handling, and access control, ensuring that only authorized users can access specific features of the system.

4.4 Event Management:

This module is responsible for creating, updating, and managing events within the system. Administrators can add event details such as title, date, time, venue, and seat availability, while also editing or deleting events when required. Students can view available events, check their status, and access detailed information. The module ensures proper organization of events and helps maintain accurate and up-to-date event data.

4.5 Event Registration and Participation:

This module enables students to register for events and manage their participation. Students can browse available events, check seat availability, and register online. The system records each registration, prevents duplicate entries, and updates seat counts automatically.

Advantages

a)Efficient Event Management: The system simplifies the process of organizing and managing events by automating registrations, updates, and tracking, reducing manual work and errors.

b) User-Friendly Interface: It provides an easy-to-use platform for both students and administrators, ensuring smooth navigation, quick access to information, and better user experience.

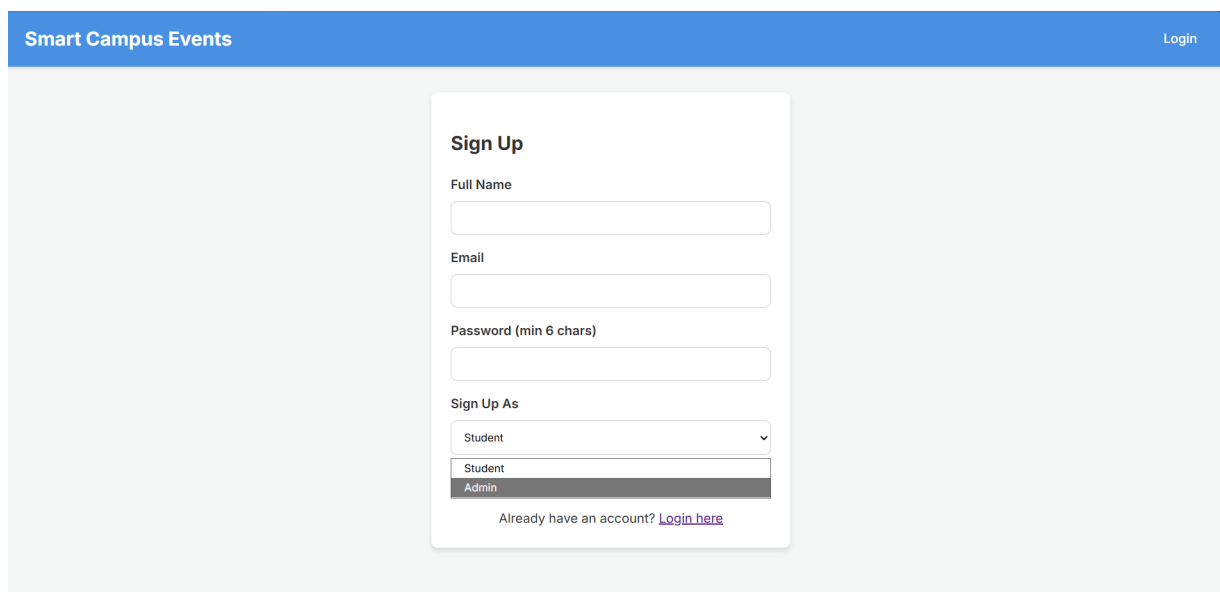
Disadvantages

The system depends on internet connectivity and requires proper maintenance of the server and database. Users with limited technical knowledge may face difficulty, and there is a risk of security issues if sensitive data is not properly protected.

Chapter 5

RESULTS AND DISCUSSIONS

The Smart Campus Event Management System was successfully implemented and tested, demonstrating efficient performance in managing events and user activities. The system allows students to register and track events easily, while administrators can manage events and monitor participation effectively. Features like secure login, OTP verification, and real-time updates functioned accurately. The results show improved organization, reduced manual effort, and better communication, making the system reliable and suitable for modern campus environments.



The screenshot displays the 'Smart Campus Events' web application interface. At the top, a blue header bar contains the text 'Smart Campus Events' on the left and a 'Login' link on the right. The main content area is light gray and features a central white 'Sign Up' form. The form includes the following elements: a title 'Sign Up', input fields for 'Full Name', 'Email', and 'Password (min 6 chars)', a 'Sign Up As' dropdown menu currently set to 'Student', and a list of roles 'Student' and 'Admin' with 'Admin' selected. At the bottom of the form, there is a link that says 'Already have an account? [Login here](#)'.

Figure 5.1: output image 1

Figure 5.1 shows the user registration page of the Smart Campus Event Management System, where new users can create an account. It allows users to enter details such as full name, email, password, and select their role (student or admin). This page ensures secure and structured user onboarding before accessing the system features.

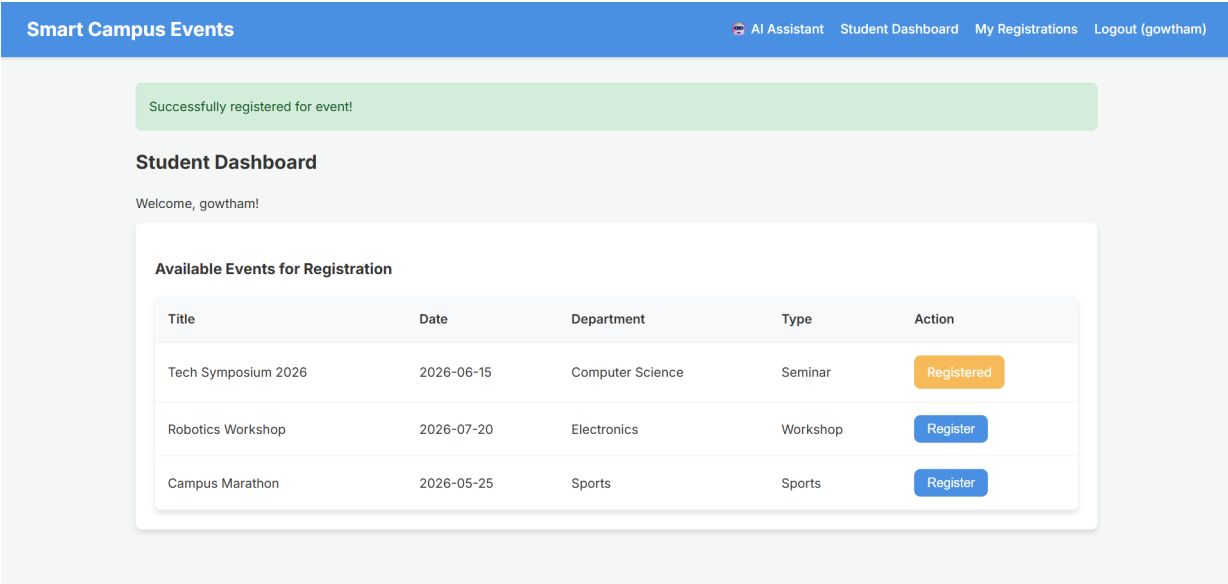


Figure 5.2: Student Dashboard

Figure 5.2 shows the student dashboard of the Smart Campus Event Management System, where users can view available events and their registration status. It displays details such as event title, date, department, and type along with options to register. The dashboard provides a user-friendly interface for managing event participation efficiently.

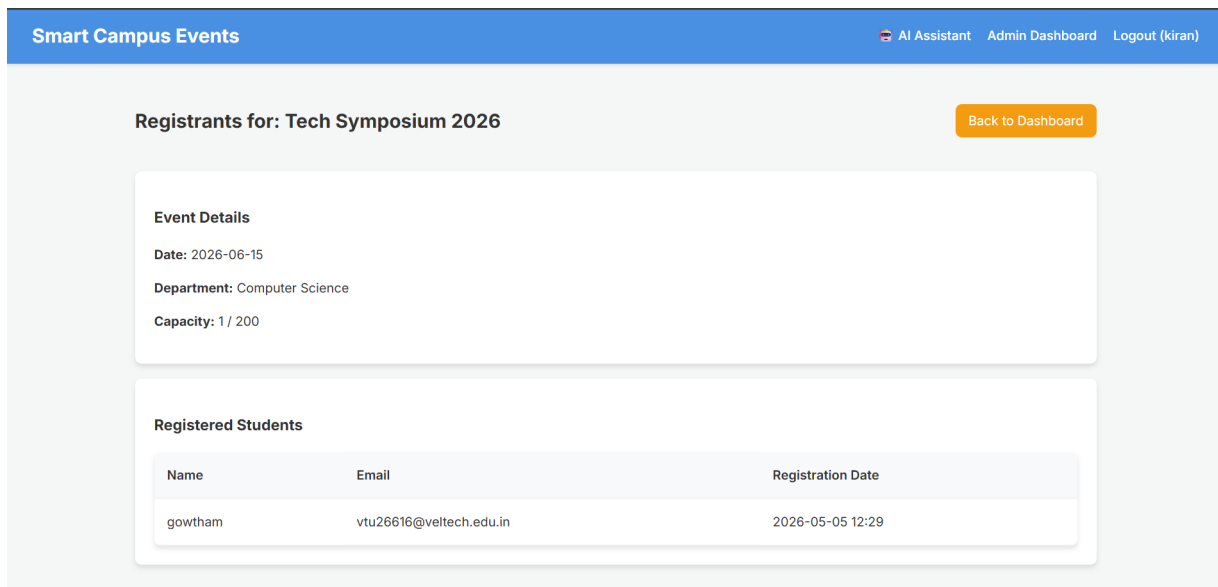


Figure 5.3: Admin Dashboard

Figure 5.3 shows the registrants view of the Smart Campus Event Management System, where administrators can view detailed information about a specific event and its participants. It displays event details such as date, department, and capacity, along with a list of registered students including their names, email addresses, and registration dates. This interface provides a clear and organized way to monitor event registrations and manage participant information efficiently.

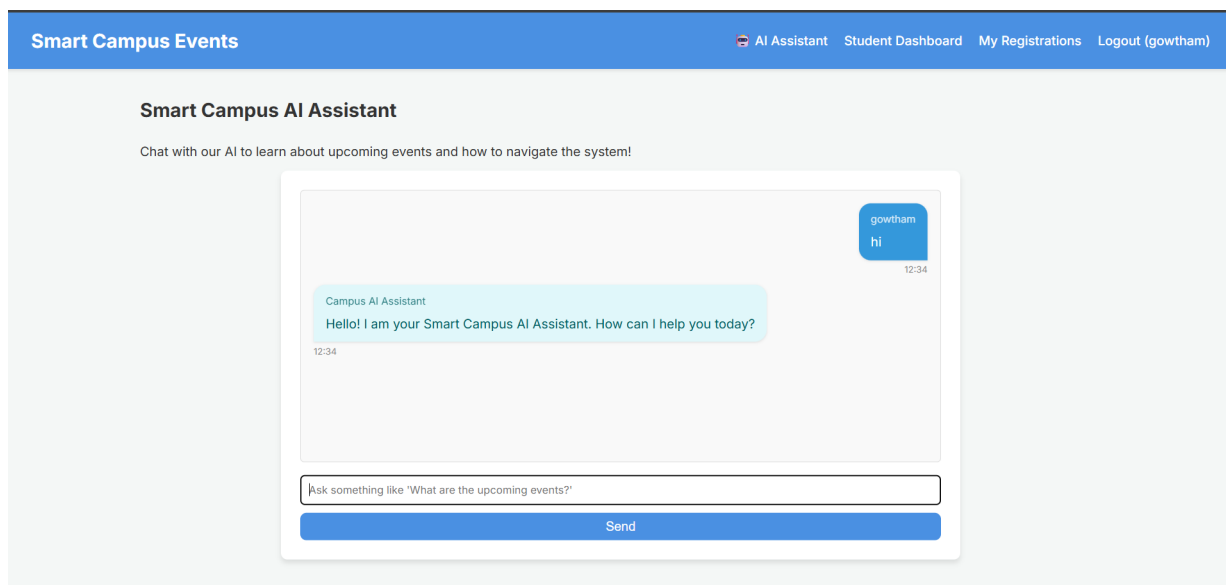


Figure 5.4: AI Assistant interface

Figure 5.4 shows the Smart Campus AI Assistant interface of the Smart Campus Event Management System, where students can interact with an AI chatbot to get information about upcoming events and system navigation. The interface includes a chat window displaying user queries and AI responses, along with a text input field for entering messages. It provides a user-friendly and interactive way for students to quickly access event-related information and assistance.

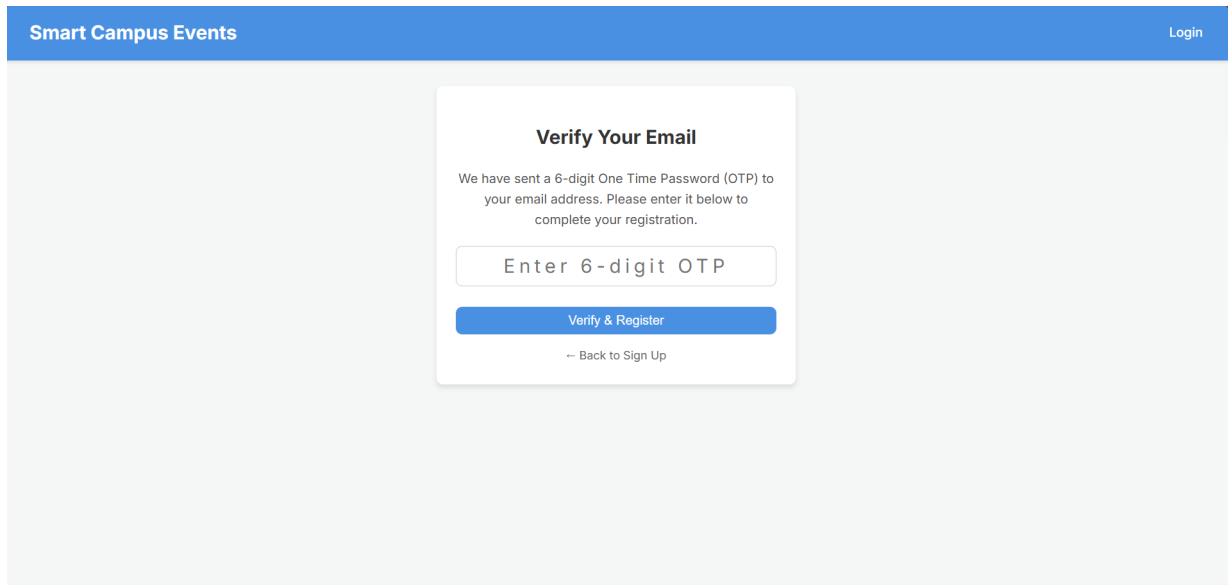
The screenshot shows a web interface for 'Smart Campus Events'. At the top, there is a blue header bar with the text 'Smart Campus Events' on the left and a 'Login' link on the right. The main content area is light gray and features a white rectangular card in the center. The card is titled 'Verify Your Email' in bold. Below the title, it says: 'We have sent a 6-digit One Time Password (OTP) to your email address. Please enter it below to complete your registration.' There is a text input field with the placeholder text 'Enter 6-digit OTP'. Below the input field is a blue button labeled 'Verify & Register'. At the bottom of the card, there is a link that says '← Back to Sign Up'.

Figure 5.5: OTP Verification

Figure 5.5 shows the email verification page of the Smart Campus Event Management System, where users confirm their registration by entering a One-Time Password (OTP) sent to their email address. The interface includes an input field for the 6-digit OTP and a “Verify Register” button to complete the process. It also provides an option to return to the sign-up page. This step ensures secure user authentication and prevents unauthorized registrations.

Chapter 6

CONCLUSION AND FUTURE ENHANCEMENTS

6.1 Conclusion

The Smart Campus Event Management System successfully addresses the challenges of managing college events by providing a simple and efficient digital solution. It streamlines event creation, registration, and participation while ensuring secure access and better communication. The system reduces manual work, improves organization, and enhances user experience for both students and administrators. Overall, it serves as a reliable and scalable platform that can be further enhanced with advanced features to meet future campus needs.

6.2 Future Enhancements

The system can be enhanced by adding a mobile application for easier access, integrating online payment for paid events, and implementing QR-based attendance tracking. Advanced analytics dashboards, push notifications, and AI-based event recommendations can further improve user experience, scalability, and efficiency for managing large-scale campus events.

Chapter 7

ADDRESSING COMPLEX ENGINEERING PROBLEM AND SDG

7.1 Mapping of the Project to Complex Engineering Problem (CEP)

Level	Attribute	Characteristics	Wks	Justification
WP1	Depth of knowledge	Requires in-depth engineering knowledge	WK3, WK4, WK5, WK6	Knowledge in Spring Boot, MVC, DB design, REST APIs, frontend tech for scalable system.
WP2	Conflicting requirements	Technical and non-technical conflicts	WK4, WK5, WK6	Balances usability with secure authentication and validation.
WP3	Depth of Analysis	Analytical and design thinking	WK3, WK4, WK5	System design, schema creation, validation, event handling.
WP4	Familiarity	Partially new problems	WK4, WK8	Real-time updates, user interaction, admin analytics.
WP5	Applicable Codes	Limited standard solutions	WK6	Custom logic for filtering, tracking, reporting.
WP6	Stakeholder Needs	Multiple stakeholders	WK5, WK6, WK7	Supports students, faculty, administrators.
WP7	Interdependence	System-level integration	WK3, WK5, WK6	Integrates UI, backend, DB, and security.

7.2 Mapping of the Project to Complex Engineering Activities (CEA)

EA No.	Attribute	Characteristics	Justification
EA1	Range of Resources	Use of diverse technologies	Uses Spring Boot, MVC, JPA, Thymeleaf, and databases for full-stack development.
EA2	Level of Interactions	Complex module interaction	Handles event management, registration, validation, and authentication modules.
EA3	Innovation	Modern technologies	Implements real-time updates and secure admin operations.
EA4	Consequences	Impact on users	Improves efficiency and reduces manual errors.
EA5	Familiarity	Beyond standard	Dynamic real-time system

7.3 SDG Mapping (CEP Section)

SDG No.	SDG Title	Relevance
SDG 4	Quality Education	Supports academic events and student learning.
SDG 9	Innovation	Promotes digital systems in education.
SDG 10	Reduced Inequalities	Ensures accessibility for all users.
SDG 11	Sustainable Communities	Encourages organized and efficient campus events.

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